BIAP Global country validation - 2026-10-16

- Code validated: `d66fc6753347e4cf7c3734044a4520d5d002fa4c` (PR #52 runtime code). Branch tip `f5b6fb48d5f276a9abdab564816d351df3838abe` only adds a CI pin (`expo-doctor@1.20.4`); no runtime change.
- Test window (UTC): 2026-10-16T01:23:57.756228+00:00 .. 2026-10-16T01:48:51.640054+00:00 (DE re-run after bounded refresh: 2026-10-16T01:48:41.641401+00:00 .. 2026-10-16T01:48:51.640054+00:00)
- CI on branch (PR #52): global-backend, backend, android-bundle all PASS; android-installable-apk skipped on PRs. Validated code SHA `d66fc6753347e4cf7c3734044a4520d5d002fa4c`; report commit is the branch tip at push time.
- Mode: `BIAP_GLOBAL_ONLINE_FIRST=1`, in-process API on an isolated copy of the cached provider data; no bulk downloads, nothing written to production data.
- Sample: top companies by market cap from each exchange's deep-scan cache (25 per market, 13 each for NASDAQ/NYSE).
- Data is real cached provider data, **not live quotes**: price age is median 1.5-2.8 days (see table).

Summary

Mkt	Status	Tested	Analyzed	Evid P/W/B	No price	Price age med/max	Calls
US	WARN	26	100%	26/0/0	0	1.5d / 2.5d	HOLD_OR_WATCH:5, NO_RECOMMENDATION:21
GB	PASS	25	100%	25/0/0	0	1.8d / 2.8d	BUY_CANDIDATE:1, HOLD_OR_WATCH:9, NO_RECOMMENDATION:15
FR	PASS	25	100%	25/0/0	0	1.8d / 2.8d	HOLD_OR_WATCH:9, NO_RECOMMENDATION:16
IT	WARN	25	100%	22/0/3	0	1.8d / 2.8d	BUY_CANDIDATE:1, HOLD_OR_WATCH:4, NO_RECOMMENDATION:20
SE	PASS	25	100%	25/0/0	0	2.8d / 2.8d	BUY_CANDIDATE:1, HOLD_OR_WATCH:7, NO_RECOMMENDATION:17
NO	WARN	25	100%	25/0/0	0	1.8d / 2.8d	BUY_CANDIDATE:10, HOLD_OR_WATCH:7, NO_RECOMMENDATION:8
DE	WARN	25	100%	23/2/0	0	2.8d / 2.8d	HOLD_OR_WATCH:2, NO_RECOMMENDATION:23

Status rule: BLOCK = >20% of sample evidence-blocked or no coverage; WARN = any evidence-BLOCK name, non-fresh peer benchmark, BUY share >30%, missing ISIN/price; else PASS.

Per-market detail

USA (NASDAQ+NYSE) - WARN

- Governance: {'REVIEW': 21, 'ACCEPT': 5}; peer-scan state: {'ok': 26}; distress: {'ELEVATED_RISK': 2, 'LOW_RISK': 16, 'INSUFFICIENT_DATA': 7, 'HIGH_RISK': 1}
- Tested at: 2026-10-16T01:23:57.756228+00:00 .. 2026-10-16T01:24:18.502394+00:00
- Missing prices: none; missing ISIN: 26; degraded providers: 0
- BUY_CANDIDATE: none; fail-closed BUY (benchmark unavailable): none
- Finding: ISIN missing for 26/26

UK (LSE) - PASS

- Governance: {'REVIEW': 15, 'ACCEPT': 10}; peer-scan state: {'ok': 25}; distress: {'LOW_RISK': 9, 'ELEVATED_RISK': 5, 'INSUFFICIENT_DATA': 8, 'HIGH_RISK': 2, 'NOT_APPLICABLE': 1}
- Tested at: 2026-10-16T01:24:19.157011+00:00 .. 2026-10-16T01:26:23.249699+00:00
- Missing prices: none; missing ISIN: 0; degraded providers: 0
- BUY_CANDIDATE: ['BHP']; fail-closed BUY (benchmark unavailable): none

France (Euronext Paris) - PASS

- Governance: {'REVIEW': 16, 'ACCEPT': 9}; peer-scan state: {'ok': 25}; distress: {'NOT_APPLICABLE': 4, 'INSUFFICIENT_DATA': 8, 'HIGH_RISK': 6, 'LOW_RISK': 4, 'ELEVATED_RISK': 3}
- Tested at: 2026-10-16T01:26:26.888843+00:00 .. 2026-10-16T01:28:45.802477+00:00
- Missing prices: none; missing ISIN: 0; degraded providers: 0
- BUY_CANDIDATE: none; fail-closed BUY (benchmark unavailable): none

Italy (Euronext Milan) - WARN

- Governance: {'ABSTAIN': 3, 'REVIEW': 17, 'ACCEPT': 5}; peer-scan state: {'ok': 25}; distress: {'ELEVATED_RISK': 2, 'NOT_APPLICABLE': 12, 'LOW_RISK': 4, 'INSUFFICIENT_DATA': 5, 'HIGH_RISK': 2}
- Tested at: 2026-10-16T01:28:47.557331+00:00 .. 2026-10-16T01:32:44.072118+00:00
- Missing prices: none; missing ISIN: 0; degraded providers: 0
- BUY_CANDIDATE: ['PIRC']; fail-closed BUY (benchmark unavailable): none
- Evidence BLOCK: ['A2A', 'MB', 'REC']
- Finding: 3/25 names evidence-BLOCK (fail-closed, correct)

Sweden (Nasdaq Stockholm) - PASS

- Governance: {'ACCEPT': 8, 'REVIEW': 17}; peer-scan state: {'ok': 25}; distress: {'LOW_RISK': 14, 'ELEVATED_RISK': 2, 'INSUFFICIENT_DATA': 5, 'NOT_APPLICABLE': 4}
- Tested at: 2026-10-16T01:32:55.856270+00:00 .. 2026-10-16T01:34:56.936762+00:00
- Missing prices: none; missing ISIN: 0; degraded providers: 0
- BUY_CANDIDATE: ['SKA.B']; fail-closed BUY (benchmark unavailable): none

Norway (Euronext Oslo) - WARN

- Governance: {'REVIEW': 8, 'ACCEPT': 17}; peer-scan state: {'ok': 25}; distress: {'NOT_APPLICABLE': 3, 'ELEVATED_RISK': 4, 'LOW_RISK': 11, 'INSUFFICIENT_DATA': 5, 'HIGH_RISK': 2}
- Tested at: 2026-10-16T01:35:00.721012+00:00 .. 2026-10-16T01:36:40.234501+00:00
- Missing prices: none; missing ISIN: 0; degraded providers: 0
- BUY_CANDIDATE: ['AKSO', 'BNOR', 'DOFG', 'EPR', 'HAUTO', 'MOWI', 'SNTIA', 'SOFF', 'SOMA', 'YAR']; fail-closed BUY (benchmark unavailable): ['MEDI']
- Finding: BUY concentration 10/25 - needs analyst review

Germany (Xetra) - WARN

- Governance: {'REVIEW': 23, 'ACCEPT': 2}; peer-scan state: {'ok': 25}; distress: {'ELEVATED_RISK': 7, 'NOT_APPLICABLE': 6, 'INSUFFICIENT_DATA': 8, 'HIGH_RISK': 1, 'LOW_RISK': 3}
- Tested at: 2026-10-16T01:48:41.641401+00:00 .. 2026-10-16T01:48:51.640054+00:00
- Missing prices: none; missing ISIN: 0; degraded providers: 0
- BUY_CANDIDATE: none; fail-closed BUY (benchmark unavailable): none
- Finding: 2/25 names evidence-WARN

Germany bounded refresh (approved 2026-10-16)

- Before: peer-scan cache stale for 25/25 (cachedAt 2026-10-14T08:21Z), all NO_RECOMMENDATION, no BUY possible.
- Action: one bounded `scan_global_market(DE, XETRA, deep_limit=25, time_budget_sec=480)` run, online-first, executed **only against the isolated temp data copy**. Production `/var/lib/biap-global` was not written (production `scan-cache/DE/XETRA.json` mtime still 2026-10-14 08:21Z).
- Scan result: status NO_RECOMMENDATION, rankingEligible true, 406 discovered / 403 screened (99.26%), fundamental coverage 98.77%, 405 deep-analyzed, 0 screening errors; run took 19 s. New cachedAt 2026-10-16T01:48:32Z.
- After: peer state ok 25/25; calls NO_RECOMMENDATION 23, HOLD_OR_WATCH 2; evidence PASS 23 / WARN 2 (CBK, MUV2) / BLOCK 0; no missing price/ISIN; no BUY.
- The refreshed file was subsequently promoted to production under a separate approval (see next section).

Temporary validation copy removal

- 29,329 files, 467 MB, no hardlinks. Compared against `/var/lib/biap-global` by SHA-256: 29,327 identical, 0 missing from production, 2 differing.
- The 2 differences carry no unique content: AAPL `latest.json` differs only in `fetchedAt` (production is newer); `isin_lei.sqlite3` has identical 189,932 rows, only `meta.builtAt` differs (copy was rebuilt at validation time).
- Deleted only `/home/ubuntu/biap-validation-20261016/data`. Kept: per-ticker results `out/`, `summary.json`, `run.log`, `tools/`, code copy `src/`, and `de-refresh/` (pre-refresh DE results + fresh XETRA.json). Production data, backups, ESEF archives, DMA agents and rollback snapshots untouched. Free disk now 8.9 GB.

Germany cache promotion (approved 2026-10-16)

- Integrity: valid JSON, schemaVersion 3, DE/XETRA, 150 deep rows, 0 rows without company; same ticker universe as previous except 7 swapped names; status NO_RECOMMENDATION, rankingEligible true.
- Provenance: produced by `scan_global_market` at code `d66fc67` from a SHA-256-verified copy of production data (online-first); sha256 `50eb260dba43....`
- Freshness: cachedAt 2026-10-16T01:48:32Z; the peer benchmark is valid for 24 h (**expires 2026-10-17T01:48Z**, then DE fails closed again until the next refresh).
- Promotion: atomic replace of `/var/lib/biap-global/scan-cache/DE/XETRA.json` (530,485 -> 544,932 bytes). Nothing else written; no service restart.
- Rollback: previous file preserved with checksum at `/var/lib/biap-global/backup-pre-de-scan-promote-20261016T015539Z/` (ROLLBACK.txt holds the copy command).
- Verified on production API 8091 after promotion: DE/SAP sector P/E 22.39 present (benchmark restored), call NO_RECOMMENDATION.

Italy evidence-BLOCK investigation (official sources, online)

Name	Official finding	Verdict
A2A (A2A S.p.A.)	GLEIF confirms LEI 81560076E3944316DB24 = A2A S.p.A. The ESEF packages filed under that issuer in OAM 1INFO carry entity LEI 815600B7FD80E48C1896 = **A2A Energia S.p.A.** (a subsidiary). filings.xbrl.org has no entity for A2A S.p.A.	Correctly BLOCK: the parent's own ESEF is not on the public aggregator; the subsidiary's figures must not be used. Needs the parent's package from OAM 1INFO / issuer IR.
MB (Mediobanca)	Official FY2025 ESEF (filed 2026-04-07/15, 0 errors) covers a **6-month transition period 2025-07-01..2025-12-31**; income-statement facts use issuer extension tags; no 12-month figures.	Correctly BLOCK: never annualized (BLOCK_NO_12_MONTH_INCOME_STATEMENT). Prior FY (to 2025-06-30) net income 1,332.5M exists but is stale. No TTM derivable.
REC (Recordati)	Official FY2025 ESEF (0 errors) contains Revenue 2,618.4M and ProfitLoss 443.6M, but the duration contexts are **tagged 2025-12-31 -> 2026-01-01 (one day)**, a filer tagging defect. Equity is an instant and is read.	Correctly BLOCK: a 1-day "annual" period is rejected. Net income corroborates against the equity-statement column (443.6M); revenue does not. Recommend asking the issuer/OAM for a corrected package; no code change.

No code was changed for Italy: all three blocks are source defects or a legitimate short period, and the engine fails closed on them.

Norway BUY_CANDIDATE audit (10 names)

All ten rest on FY2025 annual figures dated 2025-12-31 (289 days old; no interim/TTM evidence for Norway), relative P/E discount to a same-sector scan median, evidence PASS (coverage 0.875-1.0, official ESEF), and decision confidence 0.56-0.67 vs accept threshold 0.55. No contradictions were recorded.

Ticker	Sector	P/E vs sector	Peers	Distress	Conf	Flags
AKSO	Energy	7.5 vs 19.5	12	ELEVATED_RISK	0.58	elevated distress, extreme discount, near threshold
BNOR	Energy	16.2 vs 19.5	12	ELEVATED_RISK	0.58	elevated distress, near threshold
DOFG	Industrials	6.9 vs 17.6	29	INSUFFICIENT_DATA	0.56	no distress assessment, extreme discount, near threshold
EPR	Cons. Discr.	17.1 vs 23.6	5	LOW_RISK	0.63	small peer group
HAUTO	Industrials	6.8 vs 17.6	29	LOW_RISK	0.63	extreme discount, fundamental vote negative (-0.05)
MOWI	Cons. Staples	12.8 vs 22.7	6	INSUFFICIENT_DATA	0.57	no distress assessment, small peer group, near threshold
SNTIA	Industrials	14.1 vs 17.6	29	ELEVATED_RISK	0.58	elevated distress, near threshold

Ticker	Sector	P/E vs sector	Peers	Distress	Conf	Flags
SOFF	Energy	4.2 vs 19.5	12	INSUFFICIENT_DATA	0.61	no distress assessment, extreme discount (likely one-off earnings)
SOMA	Energy	6.3 vs 19.5	12	INSUFFICIENT_DATA	0.61	no distress assessment, extreme discount, fundamental vote negative (-0.05)
YAR	Materials	7.2 vs 23.4	4	LOW_RISK	0.67	small peer group, extreme discount

Assessment (counts): 3 BUYs carry ELEVATED_RISK distress; 4 have no distress assessment; 3 rest on peer groups of <=6; 6 trade at <40% of the sector P/E (typical of one-off or cyclical earnings); 2 have a non-positive fundamental vote and are carried only by the relative-P/E comparison; every one uses 289-day-old annual earnings. Only **EPR, HAUTO, YAR** carry LOW_RISK distress, and each still has one of the flags above. **None of the 10 should be presented as a clean BUY.**

Governance rule - approved and implemented globally (commit 4a906e9, all markets via `service._analysis_payload`): BUY_CANDIDATE is routed to REVIEW when (1) distress is not LOW_RISK (NOT_APPLICABLE is allowed; unknown/insufficient data counts as unverified), (2) the same-sector peer group has fewer than 8 valid companies, or (3) relative P/E is the only positive signal. Applying the rule to the table above, all 10 names route to REVIEW (AKSO/BNOR/SNTIA/DOFG/MOWI/SOFF/SOMA by distress, EPR/MOWI/YAR by peer count, HAUTO and SOMA by P/E-only); this is derived from the audit table, not a live re-scan. Regression coverage: `analysis/tests/test_buy_governance_rules.py`, 236 parametrised cases across every market. Interim/TTM evidence for Norway remains future work.

USA ISIN resolution (revised 2026-10-16)

- **Superseded approach:** the earlier SEC 13(f) CUSIP -> ISO 6166 derivation is removed (commit 4a906e9). CUSIPs are CGS-licensed, and no restricted CUSIP dataset is used without confirmed licensing.
- **Current resolver (`analysis/global_markets/us_isin.py`):** legally open sources only - GLEIF (CC0) and OpenFIGI. Flag-gated, best effort, pure functions, nothing persisted. Ambiguity returns no ISIN.
- **Missing ISIN never invalidates analysis:** a US equity with no ISIN is still analysed from SEC filings and market data; the ISIN is reported as unresolved only. The earlier "25/26 resolved" figure came from the removed CUSIP chain and is no longer a claim of this report.
- Status: resolver and tests merged on the PR #52 branch; production wiring and live coverage measurement are pending deploy approval.

Expo CI failure

- Root cause: unpinned `npx expo-doctor` resolved 1.21.4, whose schema check rejects `ios.icon` pointing at a PNG (`mobile/app.json`).
- Isolated, deploy-safe fix on PR #52: workflow pins `expo-doctor@1.20.4` (check still enforced).
- Real fix prepared in separate draft PR #54 (removes the redundant `ios.icon`): its CI runs unpinned expo-doctor and **passes** (android-bundle, typecheck). Merging it to main triggers Expo EAS build and the backend deploy workflow (`mobile/**`), so it is held for an approved deploy window.

Remaining failures and warnings

1. **Italy**: A2A needs the parent's own ESEF from OAM 1INFO; Recordati needs a corrected package from the issuer; Mediobanca stays blocked until a 12-month statement exists.
2. **Germany**: production benchmark fresh only until 2026-10-17T01:48Z; CBK and MUV2 evidence WARN. The first controlled refresh was run 2026-10-16T02:50Z (see below). A lightweight 24 h online-first benchmark refresh is implemented (`benchmark_refresh.py`; atomic, validated, keeps one previous generation, failed refresh leaves the cache untouched and fails closed). The systemd timer units are in `deploy/systemd/` but NOT installed.
3. **Norway**: the 10 flagged BUYs now route to REVIEW under the implemented rule (live re-scan pending deploy).
4. **USA**: licence-clean resolver (GLEIF + OpenFIGI) implemented, flag-gated; live ISIN coverage not yet measured.
5. **CI/deploy**: PR #54 must wait for a deploy window; PR #52's CI relies on the doctor pin until then. Italy A2A parent ESEF and Recordati corrected filing remain unresolved; evidence BLOCK retained.
6. Sample is top-by-market-cap from cached data (prices 1-3 days old), not live quotes or the full universe.

Decision-stability evidence

- 8 regression tests (`analysis/tests/test_global_decision_stability.py`): missing/stale/expired cache never yields BUY; fresh-to-stale never flips NO_RECOMMENDATION to BUY; contradiction preserved while fresh; legitimate discount BUY still allowed; identical evidence gives identical decisions.
- Full backend suite re-run 2026-10-16 on tip 4a906e9 (`pytest analysis/tests`, online-first flag off as in CI): **1439 passed, 0 failed**. Targeted run of governance, decision-stability, service/source-safety, decision-support, US-ISIN and benchmark-refresh tests: 279 passed.
- Test-isolation caveat (resolved, see "Controlled Germany refresh" below): with `BIAP_GLOBAL_ONLINE_FIRST=1` exported, 35 disk-persistence tests used to fail; `tests/conftest.py` now pins persistence mode per test.
- CI on 4a906e9 (PR #52): BIAP Global check and Release candidate check green. PR #54 (Expo doctor) CI green. Nothing merged, nothing deployed.
- Live example: Norway MEDI (P/E 32.1, no sector benchmark) -> REVIEW / NO_RECOMMENDATION "valuation benchmark unavailable".

Controlled Germany refresh (2026-10-16, manual, single run)

Command: `BIAP_GLOBAL_ONLINE_FIRST=1 python -m global_markets.benchmark_refresh DE XETRA` against `/var/lib/biap-global`, 17 s, status **OK**. No timer installed, no deploy, 8091 not restarted.

Item	Before	After
`scan-cache/DE/XETRA.json` cachedAt	2026-10-16T01:48:32Z (interactive)	2026-10-16T02:50:11Z (benchmark-refresh); 24 h expiry moves to 2026-10-17T02:50Z
deep rows / screening coverage / rankingEligible	150 / 99.26% / true	150 / 99.26% / true
Analysis ledger (405 analysed, evidence PASS 372 / WARN 17 / BLOCK 16, BLOCKED 11, buyQualified 0)	identical	identical
Calls + evidence status, 146 shared tickers	-	**0 changes**; distribution 118 NO_RECOMMENDATION/PASS, 16 /BLOCK, 12 / WARN, 4 HOLD_OR_WATCH/PASS; no BUY
Prices and filing periods, shared tickers	-	0 differences (no new prices fetched; existing data only)
Storage	-	+575,090 bytes = new XETRA.json (+1.5 KB) and the single `.prev` rollback generation (544,932 B); no other file created or modified

Findings to be aware of (not defects, not hidden): - The 150-row ranked sample changed membership by 4 names: out ADS, BAS, ENR, IFX; in BMW, DTG, HEI, VOW3 (all NO_RECOMMENDATION). Scores of 12 shared names moved (e.g. SAP 0.192 -> 0.096, SIE 0.207 -> 0.241) with identical prices, i.e. the refresh re-ranks peers; calls did not change. The ranking gate count WATCHLIST_HOLD went 5 -> 4 and Top-10 swapped ENR/RHM/SAP order; ENR's own new call was not individually re-analysed. - `batchLimit` is no longer present in the payload (interactive-scan field only). - Rollback: `mv XETRA.json.prev XETRA.json`; pre-refresh copy also kept outside the data dir. - Not done: a full per-company re-analysis of the 25 DE names through `/global/analyze` after the refresh. A disk-quota limit prevented building an isolated data copy; running it against the live data dir was rejected to keep historical datasets untouched.

Persistence tests vs online-first

All 35 failures were persistence-mode tests (market/universe/fundamentals caches, history store, FX, AMF, Brreg, exchange shares, etc.) that assert files are written and inherited the ambient `BIAP_GLOBAL_ONLINE_FIRST=1`. Fix (no test disabled or weakened): `tests/conftest.py` autouse fixture `_persistence_mode_by_default` removes the flag per test; `tests/test_global_online_first.py` keeps the explicit online-first assertions (it sets the flag itself) and gains a leak test. Result: **1440 passed, 0 failed with the flag exported to 1 and with it unset.**

Italy follow-up

- A2A, Recordati, Mediobanca: evidence BLOCK retained (no new official filing available; unchanged cause analysis above).
- Other 22 Italian names (2026-10-16 run on d66fc67, before the BUY-governance rule): 17 NO_RECOMMENDATION/PASS, 4 HOLD_OR_WATCH/PASS, 1 BUY_CANDIDATE/PASS (Pirelli, PIRC:

distress LOW_RISK, but its comparison agent had no valid P/E). Under the new rule PIRC needs a verified same-sector peer group of at least 8; this was not re-run after the rule landed, so its post-rule call is **unconfirmed** and must be checked after deploy.